TSM 099 Polynomials Test

Practice 3 | Version 202610122020.21

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
1. Factor completely over the integers, or write prime. $(a)\quad 2x^2+x-6$ $(b)\quad 2x^2+x+2$	1. (a) 10 pts (b) 10 pts
2. Factor completely. $(a)\quad x^2+2x-24$ $(b)\quad 9x^2-16$	2. (a) 10 pts (b) 10 pts
3. Factor using the ac-method and grouping. Show ac, the split middle term, and the grouping. $15x^2-29x+12$	3. ac; pair 4 pts Split middle term 4 pts Grouping 6 pts Factored form 6 pts
4. Multiply and simplify. $(4x-3)(5x+2)$	4. Polynomial 20 pts
5. Factor out the greatest common factor. $8x^4-4x^3+20x^2$	5. Factored form 20 pts

Instructor answer key: TSM 099 Polynomials Test

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. (a)

$(x+2)(2x-3)$

10 pts

(b)

Prime

10 pts

2. (a)

$(x-4)(x+6)$

10 pts

(b)

$(3x-4)(3x+4)$

10 pts

3. ac; pair

$ac=180;\ \ -9,\ \ -20$

4 pts

Split middle term

$15x^2-9x-20x+12$

4 pts

Grouping

$3x(5x-3)+\ -4(5x-3)$

6 pts

Factored form

$(3x-4)(5x-3)$

6 pts

4. Polynomial

$20x^2-7x-6$

20 pts

5. Factored form

$4x^2(2x^2-x+5)$

20 pts